Goal: Solve optimization problems involving volume, surface area, and distances.

Example 1. We want to construct a box with a square base and no top. If the volume is 500 cubic inches, then what is the minimum amount of material required?



Obj: Minimize $A_{(x,h)} = x^2 + 4xh$

Given: $x^2 h = 500$

$h = \frac{500}{x^2}$

$A(x) = x^2 + 4x\left(\frac{500}{x^2}\right)$

$A(x) = x^2 + \frac{2000}{x}$

$A'(x) = 2x - \frac{2000}{x^2}$ set 0

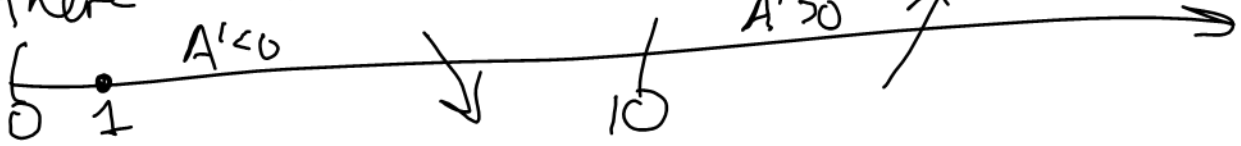
$2x = \frac{2000}{x^2}$

$2x^3 = 2000$

$x^3 = 1000$

$\boxed{x = 10}$

Verify there is a min at $x = 10$:



Rel. Min at $x=10$ + Only 1 crit. num $\Longrightarrow$ Abs. min at $x=10$

Find the surface when $x = 10$:

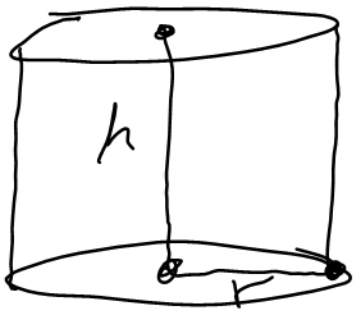
$$A(x) = x^2 + \frac{2000}{x}$$

$$A(10) = 10^2 + \frac{2000}{10}$$

$$= 100 + 200 = 300 \text{ in}^2$$

Example 2 (Ideal Soup Can)**.** A company is designing a cylindrical soup can to hold $250\pi \approx 785.4\text{cm}^3$ of liquid. What should the dimensions be to use the least amount of material?

Example 3 (Ideal Soup Can II)**.** Due to the manufacturing process, each can has to use $294\pi \approx 923.63\text{cm}^2$ of aluminum. Find the largest amount of soup these cans could store.





Obj: Max $V_{(cin)} = \pi r^2 h$

Given: $2\pi r^2 + 2\pi r h = 294\pi$

$2\pi r^2 + 2\pi r h = 294\pi$

$2\pi r h = 294\pi - 2\pi r^2$

$h = \dfrac{294\pi - 2\pi r^2}{2\pi r}$

$h = \dfrac{147}{r} - r$

$V(r) = \pi r^2 \left(\dfrac{147}{r} - r\right)$

$V(r) = 147\pi r - \pi r^3$

$V'(r) = 147\pi - 3\pi r^2 \overset{\text{set}}{=} 0$

$147\pi = 3\pi r^2$

$49 = r^2$

$r = 7$

Verify its a max by the 2nd $\frac{d}{dx}$ test:

$V''(r) = -6\pi r$

$V''(7) = -42\pi < 0$ ∩

Find the maximum volume: Find $V(7)$

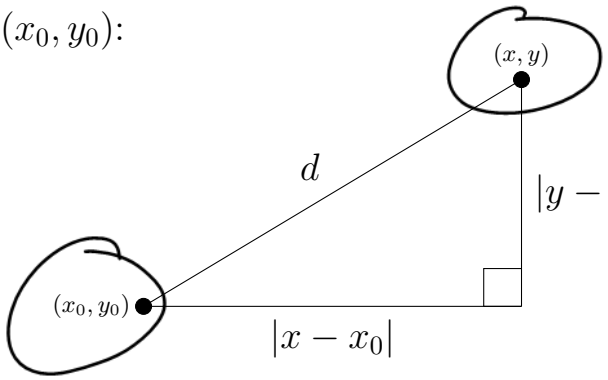
$$V(r) = \pi r^2\left(\frac{147}{r} - r\right)$$

$$V(7) = 49\pi\left(\frac{147}{7} - 7\right)$$

$$= 686\pi \approx 2155.13 \text{ cm}^3$$

Distances

Recall the formula for computing the distance between the points (x, y) and (x_0, y_0):



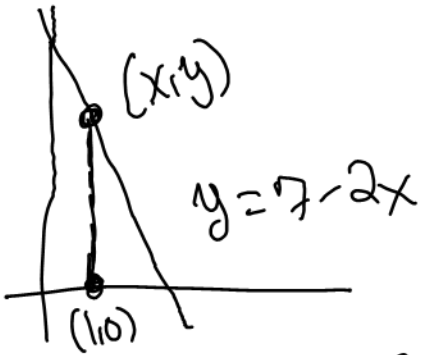
$$d = \sqrt{(x - x_0)^2 + (y - y_0)^2}$$

Since distances are positive, squaring and taking square roots preserve inequality. I.e., if there is a minimum distance:

$$[\text{minimum}]^2 \leq [d]^2$$
$$\sqrt{[\text{minimum}]^2} \leq \sqrt{[d]^2}$$
$$\text{minimum} \leq d$$

In short: Finding the location where the optimal distance occurs is the same as finding the location where the squared distance occurs.

Example 4. What is the minimum distance between $y = 7 - 2x$ and the point $(1, 0)$?



$$d = \sqrt{(x-1)^2 + (y-0)^2}$$
$$= \sqrt{(x-1)^2 + y^2}$$

Obj: Min $S(x,y) \stackrel{\text{def}}{=} [d(x,y)]^2 = (x-1)^2 + y^2$

Given: $y = 7 - 2x$

$$S(x) = (x-1)^2 + (7-2x)^2$$
$$= x^2 - 2x + 1 + 49 - 28x + 4x^2$$
$$S(x) = 5x^2 - 30x + 50$$

$$S'(x) = 10x - 30 \overset{\text{set}}{=} 0$$
$$10x = 30$$
$$x = 3$$

Let's go ahead and find the y-coordinate

$$y = 7 - 2x$$
$$y = 7 - 2(3) = 7 - 6 = 1$$

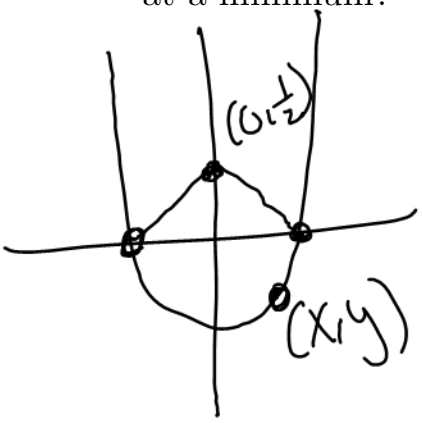
Ie, the minimum occurs at $(3,1)$

$$S(3,1) = (3-1)^2 + 1^2 = 5$$

To find the minimum distance, if is

$$\sqrt{S(3,1)} = \boxed{\sqrt{5}}$$

Example 5. Where is the distance between $(0, \frac{1}{2})$ and the parabola $y = x^2 - 1$ at a minimum?



Obj: Min

$S(x,y) = [d(x,y)]^2 = x^2 + (y - \frac{1}{2})^2$

Given: $y = x^2 - 1$

$S(x) = x^2 + (x^2 - 1 - \frac{1}{2})^2$

$= x^2 + (x^2 - \frac{3}{2})^2$

$S(x) = x^4 - 2x^2 + \frac{9}{4}$

$S'(x) = 4x^3 - 4x$ set 0

$= 4x(x-1)(x+1) = 0$

$\Rightarrow x = -1, 0, 1$

$S''(x) = 12x^2 - 4$

$S''(-1) = S''(1) > 0$

$S''(0) < 0$

